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Saijo; Ryohei et al.

Information processing apparatus, information processing method, and program for placing an agent in a virtual space

Abstract

In accordance with an aspect of the present invention, an information processing apparatus includes an interface that transmits or receives data to or from an input apparatus that inputs user position information indicating a position of a user, and a processor that calculates an evaluation value for an evaluation target position based on the user position information, determines a position in a virtual space or a real space where an agent supporting the user is placed based on the evaluation value, and places the agent at the determined position in the virtual space or real space.

Inventors: Saijo; Ryohei (Musashino, JP), Watanabe; Masahiro (Musashino, JP), Sato; Tae (Musashino, JP)
Applicant: Nippon Telegraph and Telephone Corporation (Tokyo, JP)
Family ID: 79267672
Assignee: Nippon Telegraph and Telephone Corporation (Tokyo, JP)
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Primary Examiner: Hsu; Joni

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage application under 35 U.S.C. § 371 of International Application No. PCT/JP2020/023792, having an International Filing Date of Jun. 17, 2020, the disclosure of which is considered part of the disclosure of this application, and is incorporated by reference in its entirety into this application.

TECHNICAL FIELD

(2) An aspect of the present invention relates to an information processing apparatus, an information processing method, and a program for placing an agent that supports a user, for example.

BACKGROUND ART

(3) In recent years, a system for placing an agent for supporting a user in a virtual space or the like has been provided. For example, such a system displays the agent on a screen that the user visually

recognizes. The system provides predetermined support to the user through an operation of the agent or the like.

CITATION LIST

Non Patent Literature

(4) NPL 1: Takahiro Tanaka et al., Acceptable Dialogue Start Supporting Agent for Multi-Tasking Online Communication, 24th Fuzzy System Symposium, DT2-3, pp. 586-589, 2008

SUMMARY OF THE INVENTION

Technical Problem

(5) It is preferable for an agent to be placed at a position that is easily recognized by a user and does not block the user from seeing what the user wants to watch. Thus, a technology for placing an agent at an appropriate position in a field of view of a user is provided.

Means for Solving the Problem

(6) In accordance with an aspect of the present invention, an information processing apparatus includes: an interface configured to transmit or receive data to or from an input apparatus configured to input user position information indicating a position of a user; and a processor configured to calculate an evaluation value for an evaluation target position based on the user position information, determine a position in a virtual space or a real space where an agent supporting the user is placed, based on the evaluation value, and place the agent at the determined position in the virtual space or real space.

Effects of the Invention

(7) In accordance with an embodiment, the agent system can place the agent at a position that is easily recognized by the user and does not block the user from seeing what the user wants to watch.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram illustrating a configuration example of an agent system according to a first embodiment.

(2) FIG. 2 is a block diagram illustrating a configuration example of the agent system according to the first embodiment.

(3) FIG. 3 is a functional block diagram of the processor according to the first embodiment.

(4) FIG. 4 is a diagram illustrating an operation example of the agent system according to the first embodiment.

(5) FIG. 5 is a diagram illustrating an operation example of the agent system according to the first embodiment.

(6) FIG. 6 is a diagram illustrating an operation example of the agent system according to the first embodiment.

(7) FIG. 7 is a diagram illustrating an operation example of the agent system according to the first embodiment.

(8) FIG. 8 is a diagram illustrating an operation example of the agent system according to the first embodiment.

(9) FIG. 9 is a diagram illustrating an operation example of the agent system according to the first embodiment.

(10) FIG. 10 is a sequence diagram illustrating an operation example of the agent system according to the first embodiment.

(11) FIG. 11 is a block diagram illustrating a configuration example of an agent system according to a second embodiment.

(12) FIG. 12 is a functional block diagram of a processor according to the second embodiment.

(13) FIG. 13 is a sequence diagram illustrating an operation example of the agent system according to the second embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(14) Hereinafter, embodiments according to the present invention will be described with reference to the drawings.

First Embodiment

(15) First, a first embodiment will be described.

(16) An agent system according to the embodiment places an agent in a virtual space by using a technology such as virtual reality (VR), augmented reality (AR), or mixed reality (MR). The agent system supports a user who is a support target, through an agent. Here, the agent system presents to the user that another user has an intention of starting communication by changing a direction of the agent. Content of the support that the agent system performs is not limited to a specific configuration.

(17) FIG. 1 illustrates a configuration example of an agent system **100** according to the embodiment. As illustrated in FIG. 1, the agent system **100** includes a headset **1**, an information processing apparatus **3**, and the like. The headset **1** and the information processing apparatus **3** are connected to each other and can communicate with each other.

(18) The headset **1** is a goggle type headset. The headset **1** has a structure that can be attached to and detached from a head of the user. The headset **1** displays an image to the user under the control of the information processing apparatus **3**. Further, the headset **1** measures a position, angle, acceleration, and the like of the head of the user.

(19) The headset **1** may be a glasses-type headset.

(20) The information processing apparatus **3** sets a virtual space. The information processing apparatus **3** generates an image visually recognized by an avatar of the user at a predetermined position in the virtual space and displays the image on the headset **1**.

(21) FIG. 2 is a block diagram illustrating a configuration example of the agent system **100**. As illustrated in FIG. 2, the headset **1** includes a display apparatus **11** and an input apparatus **12**. The display apparatus **11** and the input apparatus **12** are connected to the information processing apparatus **3**.

(22) The display apparatus **11** displays an image to the user under the control of the information processing apparatus **3**. Here, the display apparatus **11** is a head-mounted display mounted on the head of the user. Further, the display apparatus **11** may display different images to left and right eyes of the user to realize stereoscopic viewing.

(23) For example, the display apparatus **11** includes a display panel configured of an organic electroluminescence (EL) display, and has a display area corresponding to a visual field range of the user.

(24) The input apparatus **12** inputs posture information indicating a posture of the user and user position information indicating a position of the user to the information processing apparatus **3**. The posture information and the user position information are used for the information processing apparatus **3** to specify a field of view of the avatar of the user in the virtual space. Here, the input apparatus **12** inputs posture information indicating, for example, an operation of the head of the user wearing the headset **1** as the posture of the user to the information processing apparatus **3**. For example, the input apparatus **12** includes a sensor that measures an angle of the headset **1**, an acceleration applied to the headset **1**, and the like. The input apparatus **12** inputs a sensor signal from the sensor to the information processing apparatus **3** as the posture information. The input apparatus **12** may input an operation signal from a controller to the information processing apparatus **3** as the posture information.

(25) Further, the user position information indicates the position of the user in the virtual space. The input apparatus **12** may acquire the user position information through a sensor. Further, the input apparatus **12** may acquire the user position information through the controller. Further, the user position information may indicate a movement amount and a movement direction of the user.

(26) Further, as illustrated in FIG. 2, the information processing apparatus **3** includes, for example, a processor **31**, a memory **32**, an input and output interface **33**, and a communication unit **34**.

(27) The processor **31**, the memory **32**, the input and output interface **33**, and the communication unit **34** are connected to each other via a data bus or the like.

(28) The processor **31** has a function of controlling an operation of the entire information processing apparatus **3**. The processor **31** may include, for example, an internal cache and various interfaces. The processor **31** realizes various processing by executing a program stored in advance in the internal memory, the memory **32**, or the like.

(29) For example, the processor **31** includes a CPU and the like. Further, the processor **31** may be configured of an application specific integrated circuit (ASIC) or the like. Further, the processor **31** may be configured of a field programmable gate array (FPGA) or the like.

(30) The memory **32** stores various types of data. For example, the memory **32** functions as a ROM, a RAM, and an NVM.

(31) For example, the memory **32** stores a control program, control data, and the like. The control program and the control data are incorporated in advance according to a specification of the information processing apparatus **3**. For example, the control program is, for example, a program that supports functions realized by the information processing apparatus **3**.

(32) Further, the memory **32** temporarily stores data or the like that is being processed by the processor **31**. Further, the memory **32** may store data necessary for execution of an application program, an execution result of the application program, and the like.

(33) The input and output interface **33** is an interface for transmitting or receiving data to or from the headset **1** wirelessly or by wire. That is, the input and output interface **33** communicates with the display apparatus **11** and the input apparatus **12** of the headset **1**.

(34) For example, the input and output interface **33** is an interface that supports a universal serial bus (USB) connection, a Bluetooth (registered trademark) connection, or the like.

(35) Further, the input and output interface **33** may include an interface for transmitting or receiving data to or from the display apparatus **11**, and an interface for transmitting or receiving data to or from the input apparatus **12**.

(36) The communication unit **34** is an interface for connection to an external apparatus. For example, the communication unit **34** connects to the external apparatus through an external network. For example, the communication unit **34** is an interface that supports a wired or wireless local area network (LAN) connection.

(37) For example, the information processing apparatus **3** is a computer such as a desktop PC or a laptop PC.

(38) The agent system **100** may include a configuration as required, in addition to the configuration as illustrated in FIGS. **1** and **2**, or have a configuration in which a specific configuration is excluded from the information processing apparatus **3**.

(39) Further, the headset **1** and the information processing apparatus **3** may be integrally formed. Further, each of the headset **1** and the information processing apparatus **3** may include a plurality of apparatus.

(40) Next, functions that are realized by the information processing apparatus **3** will be described. The functions realized by the information processing apparatus **3** are realized by the processor **31** executing a program stored in the internal memory, the memory **32**, or the like.

(41) FIG. **3** illustrates the functions that are realized by the processor **31**. As illustrated in FIG. **3**, the processor **31** realizes an operation control unit **311**, a position calculation unit **312**, a virtual space control unit **313**, and the like.

(42) The virtual space control unit **313** sets the virtual space. The virtual space control unit **313** controls a size and shape of the virtual space. Further, the virtual space control unit **313** places the avatar of the user, the agent, or other object in the virtual space. The virtual space control unit **313** may place objects under the control of an external apparatus that communicates through the communication unit **34**.

(43) Further, the virtual space control unit **313** generates a field of view (an image seen from a viewpoint of the user) of the user (the avatar of the user) in the virtual space. The virtual space

control unit **313** generates a vector indicating a line of sight of the user based on the posture information from the input apparatus **12**. The virtual space control unit **313** causes the display apparatus **11** to display an image of the field of view of the user based on the vector, the user position information, and the like.

(44) Further, the operation control unit **311** controls an operation of the agent in the virtual space. Here, the operation control unit **311** causes the agent to perform an operation of supporting the user.

(45) FIG. **4** illustrates the operation of the agent in the virtual space. In an example illustrated in FIG. **4**, an avatar **201**, an agent **202**, and a speaker **203** are placed in the virtual space.

(46) The avatar **201** is an avatar that is operated by the user of the agent system **100**. The virtual space control unit **313** causes the display apparatus **11** to display a field of view of the avatar **201**.

(47) The speaker **203** is one of objects placed in the virtual space. The speaker **203** performs communication with the avatar **201**. The speaker **203** may be an object operated by a program or an avatar operated by another user.

(48) The agent **202** is an agent that supports the avatar **201** (that is, the user). The agent **202** is one of the objects placed in the virtual space.

(49) The agent **202** informs the avatar **201** that the speaker **203** has an intention of starting communication with the avatar **201**, by a direction of the agent **202**. Here, the operation control unit **311** calculates an angle p formed by a vector indicating a current direction of the agent **202** and a vector connecting the agent **202** to the speaker **203**. When the angle p is calculated, the operation control unit **311** changes a direction of the agent **202** by the angle p . That is, the operation control unit **311** rotates the agent **202** to be directed to the speaker **203**.

(50) The position calculation unit **312** determines a position of the agent **202** in the virtual space.

(51) FIG. **5** illustrates a virtual space **200** set by the virtual space control unit **313**. The virtual space **200** is X in horizontal width by Y in height (depth). In FIG. **5**, objects **204** to **206** are further placed.

(52) The position calculation unit **312** sets an agent placement area **210** in which the agent **202** can be placed. That is, the position calculation unit **312** determines a position in the agent placement area as the position of the agent **202**.

(53) The agent placement area **210** is an area divided in an $M \times N$ grid pattern in which the avatar **201** is allocated at a center of a lower end. i and j are subscripts indicating coordinates of a grid point of the agent placement area **210** in a relative coordinate system in which the avatar **201** is allocated at a center. i indicates an x coordinate, and j indicates a y coordinate.

(54) The position calculation unit **312** determines the position of the agent **202** based on an evaluation value $V_{sub.i,j}$ of the grid point (an evaluation target position). Here, the position calculation unit **312** determines a grid point at which the evaluation values $V_{sub.i,j}$ is maximized as the position of the agent. Further, when there are a plurality of grid points at which the evaluation values $V_{sub.i,j}$ is maximized, the position calculation unit **312** selects one grid point from the plurality of grid points. For example, the position calculation unit **312** may select a grid point farther from an end of the virtual space **200**. Further, the position calculation unit **312** may be set by the user as to which of left and right or upper and lower grid points is preferentially selected.

(55) The evaluation value $V_{sub.i,j}$ is calculated by the following equation.

$$V_{ij} = \begin{cases} 0 & (i_{ij} \leq 0 \text{ or } i_{ij} \geq X \text{ or } j_{ij} \leq 0 \text{ or } j_{ij} \geq Y) \\ w_1 f_1(l_{ij}) + w_2 f_2(i_{ij}) + w_3 f_3(b_{ij}) & (0 < i_{ij} < X \text{ and } 0 < j_{ij} < Y) \end{cases} \quad (V_{ij} \geq 0) \quad [\text{Math. 1}]$$

(57) $\alpha_{sub.i,j}$ is a distance between a grid point (i, j) and a left end of the virtual space **200**. Further, $\beta_{sub.i,j}$ is a distance between the grid point (i, j) and a lower end of the virtual space **200**. That is, $V_{sub.i,j}$ become 0 when the corresponding grid point is outside the virtual space **200**. The position calculation unit **312** calculates $\alpha_{sub.i,j}$ and $\beta_{sub.i,j}$, as follows.

(58) First, the position calculation unit **312** converts the grid point (i, j) into coordinates (I, J) in a coordinate system (a world coordinate system) representing the entire virtual space according to the

following equation.

$$(59) \begin{pmatrix} I \\ J \end{pmatrix} = \begin{pmatrix} \cos & -\sin \\ \sin & \cos \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad [\text{Math. 2}]$$

(60) I indicates an x coordinate in the world coordinate system, and J indicates a y coordinate in the world coordinate system. $x_{\text{sub}.0}$ indicates an x coordinate of the avatar **201** in the world coordinate system. $y_{\text{sub}.0}$ indicates a y coordinate of the avatar **201** in the world coordinate system. ϕ indicates a rotation angle of the relative coordinate system on the world coordinate system.

(61) The position calculation unit **312** calculates d_i, j based on a coordinate I and an x coordinate at the left end of the virtual space **200**. The position calculation unit **312** calculates $\beta_{\text{sub}.i,j}$ based on a coordinate J and a y coordinate at the lower end of the virtual space **200**.

(62) Next, $l_{\text{sub}.i,j}$, $\theta_{\text{sub}.i,j}$ and $b_{\text{sub}.i,j}$ will be described.

(63) FIG. **6** is a diagram illustrating $l_{\text{sub}.i,j}$, $\theta_{\text{sub}.i,j}$ and $b_{\text{sub}.i,j}$. As illustrated in FIG. **6**, $l_{\text{sub}.i,j}$ indicate a distance between the grid point (i, j) and the avatar **201**. Here, coordinates of the avatar **201** are a grid point (M/2, 1). That is, $l_{\text{sub}.i,j}$ are a distance between the grid point (i, j) and the grid point (M/2, 1).

(64) $\theta_{\text{sub}.i,j}$ is an angle formed by a vector indicating a line-of-sight direction of the avatar **201** and a vector connecting the avatar **201** to the grid point (i, j). Here, the line-of-sight direction of the avatar **201** is the y-axis direction (an up direction).

(65) $b_{\text{sub}.i,j}$ is a density of objects in a range of a predetermined radius r from the grid point (i, j).

(66) Further, $f_{\text{sub}.1}$ is a function for calculating the evaluation value based on $l_{\text{sub}.i,j}$. $f_{\text{sub}.2}$ is a function for calculating the evaluation value based on $\theta_{\text{sub}.i,j}$. $f_{\text{sub}.3}$ is a function for calculating the evaluation value based on $b_{\text{sub}.i,j}$.

(67) Further, $W_{\text{sub}.1}$, $W_{\text{sub}.2}$, and $W_{\text{sub}.3}$ are weighting coefficients of $f_{\text{sub}.1}$, $f_{\text{sub}.2}$, and $f_{\text{sub}.3}$, respectively.

(68) Next, $f_{\text{sub}.1}$ will be described.

(69) FIG. **7** is a diagram illustrating $f_{\text{sub}.1}$.

(70) $f_{\text{sub}.1}$ is a quadratic function maximized when a value at a center between a lower limit d and an upper limit D of $l_{\text{sub}.i,j}$ is input. The lower limit d may be a lower limit of $l_{\text{sub}.i,j}$ of the grid point in the field of view of the avatar **201**. Further, the upper limit D may be an upper limit of $l_{\text{sub}.i,j}$ of the grid point in the field of view of the avatar **201**.

(71) That is, $f_{\text{sub}.1}$ is the following equation.

$$(72) f_1(l_{ij}) = -\frac{1}{(\frac{d+D}{2})^2} (l_{ij} - \frac{d+D}{2})^2 + 1 (d < l_{ij} \leq D) \quad [\text{Math. 3}]$$

(73) $f_{\text{sub}.1}$ may be a function of a normal distribution maximized when the value at the center of the lower limit d and the upper limit D is input. A configuration of $f_{\text{sub}.1}$ is not limited to a specific configuration.

(74) Next, $f_{\text{sub}.2}$ will be described.

(75) FIG. **8** is a diagram illustrating $f_{\text{sub}.2}$.

(76) $f_{\text{sub}.2}$ is a quadratic function maximized when a value at a center of ϕ' and ϕ is input.

(77) As illustrated in FIG. **8**, ϕ' is an angle from a vector indicating the line of sight of the user to an end of a central visual field (an effective visual field). Further, ϕ is an angle from the vector indicating the line of sight of the user to an end of the visual field.

(78) That is, $f_{\text{sub}.2}$ is the following equation.

$$(79) f_2(\phi_{ij}) = -\frac{1}{(\frac{\phi'+\phi}{2})^2} (\phi_{ij} - \frac{\phi'+\phi}{2})^2 + 1 (\phi' < \phi_{ij} \leq \phi) \quad [\text{Math. 4}]$$

(80) $f_{\text{sub}.2}$ may be a function of a normal distribution maximized when the value at the center of ϕ' and ϕ is input.

(81) Further, $f_{\text{sub}.2}$ may be a function that increases in proportion to $\theta_{\text{sub}.i,j}$ approaching q .

(82) A configuration of f.sub.2 is not limited to a specific configuration.

(83) Next, f.sub.3 will be described.

(84) FIG. 9 is a diagram illustrating f.sub.3.

(85) f.sub.3 is a function that is inversely proportional to a density b.sub.i,j of the objects in the range of the radius r from the grid point (i, j).

(86) f.sub.3 is the following equation.

$$b_{ij} = \frac{a_{ij}}{2r^2}$$

(87)
$$f_3(b_{ij}) = \begin{cases} 1(b_{ij} = 0) & [\text{Math. 5}] \\ \frac{1}{b_{ij}}(b_{ij} > 0) \end{cases}$$

(88) a.sub.i,j is the number of objects in the range of the radius r from the grid point (i, j). In the example illustrated in FIG. 9, a.sub.i,j is 1.

(89) f.sub.3 may be a function that decreases in proportion to the density b.sub.i,j of the objects.

(90) A configuration of f.sub.3 is not limited to a specific configuration.

(91) The position calculation unit **312** outputs agent position information indicating the determined position to the virtual space control unit **313**.

(92) Next, an operation example of the agent system **100** will be described.

(93) FIG. 10 is a sequence diagram illustrating the operation example of the agent system **100**.

(94) Here, it is assumed that the speaker intends to start communication with the avatar **201**.

(95) First, the virtual space control unit **313** outputs speaker position information indicating a position of the speaker to the operation control unit **311** (S11).

(96) The operation control unit **311** changes the direction of the agent **202** based on the speaker position information from the virtual space control unit **313** (S12). When the direction of the agent **202** is changed, the operation control unit **311** performs control for causing the display apparatus **11** to display the image in which the direction of the agent **202** has been changed (S13).

(97) The display apparatus **11** updates the display of the agent **202** according to the control of the operation control unit **311** (S14).

(98) Here, the input apparatus **12** acquires the posture information of the user and the user position information (S15). When the posture information and the user position information are acquired, the input apparatus **12** inputs the acquired posture information and user position information to the information processing apparatus **3** (S16).

(99) When the input apparatus **12** inputs the posture information and the user position information, the virtual space control unit **313** outputs line-of-sight information indicating the vector indicating the line-of-sight of the user, the position of the user, and the like, object information indicating objects installed in the virtual space **200**, positions of the objects, and the like, grid point information indicating a grid point in the relative coordinate system, and the like to the position calculation unit **312** (S17).

(100) The position calculation unit **312** determines the position of the agent **202** based on the information from the virtual space control unit **313** (S18). When the position of the agent **202** is determined, the position calculation unit **312** outputs agent position information indicating the position of the agent **202** to the virtual space control unit **313** (S19).

(101) The virtual space control unit **313** changes the position of the agent **202** to the position indicated by the agent position information from the position calculation unit **312** (S20). When the position of the agent **202** is changed, the virtual space control unit **313** performs control for causing the display apparatus **11** to display an image in which the position of the agent **202** has been changed (S21).

(102) The display apparatus **11** updates the display of the agent **202** according to the control of the operation control unit **311** (S22).

(103) The agent system **100** may return to S11 or S15.

(104) Further, the virtual space control unit **313** may execute **S13** based on the information from the operation control unit **311**.

(105) The agent system configured as described above places an agent that supports the avatar of the user in the virtual space. The agent system places the agent at a position easily visible to the user while avoiding an area watched by the user based on the evaluation value of the grid point.

(106) As a result, the agent system can reduce a trouble of the user searching for the agent by placing the agent in the field of view of the user. Further, the agent system can support the user without blocking the activity of the user by placing the agent while avoiding an area watched by the user.

Second Embodiment

(107) Next, a second embodiment will be described.

(108) An agent system according to the second embodiment differs from that according to the first embodiment in that an agent that supports a user is placed in a real space. Thus, other respects are denoted by the same reference numerals and detailed description thereof will be omitted. An agent according to the second embodiment is a robot or the like.

(109) FIG. **11** is a block diagram illustrating a configuration example of an agent system **100'** according to the second embodiment. As illustrated in FIG. **11**, the agent system **100'** includes, for example, an information processing apparatus **3**, an operation apparatus **41**, and an input apparatus **42**. The operation apparatus **41** and the input apparatus **42** are connected to the information processing apparatus **3**.

(110) The operation apparatus **41** controls the operation of the agent placed in the real space. The operation apparatus **41** moves the agent according to the control of the information processing apparatus **3**. Further, the operation apparatus **41** changes the posture, direction, or the like of the agent according to the control of the information processing apparatus **3**.

(111) The input apparatus **42** detects the position, posture, and the like of the user. For example, the input apparatus **42** detects, for example, the movement of the head as the posture of the user. The input apparatus **42** inputs the posture information indicating the posture of the user and the user position information indicating the position of the user to the information processing apparatus **3**.

(112) Further, the input apparatus **42** may receive an input such as an operation of the user.

(113) The agent system **100'** may include a configuration as required, in addition to the configuration as illustrated in FIG. **11**, or have a configuration in which a specific configuration is excluded from the information processing apparatus **3**.

(114) Next, functions that are realized by the information processing apparatus **3** will be described. The functions realized by the information processing apparatus **3** are realized by the processor **31** executing a program stored in the internal memory, the memory **32**, or the like.

(115) FIG. **12** illustrates the functions that are realized by the processor **31**. As illustrated in FIG. **12**, the processor **31** realizes an operation control unit **411**, a position calculation unit **412**, a spatial information management unit **413**, an environmental information acquisition unit **414**, and the like.

(116) The spatial information management unit **413** manages information in a predetermined real space in which the agent is operated. The spatial information management unit **413** has information such as the size of the space or the placement of objects in the space. Further, the spatial information management unit **413** manages the position of the speaker.

(117) The environmental information acquisition unit **414** acquires information on surroundings in which the agent exists. Further, the environmental information acquisition unit **414** acquires the posture information and the user position information from the input apparatus. Further, the environmental information acquisition unit **414** acquires an intention of speaking of the speaker from the input apparatus.

(118) The operation control unit **411** controls an operation of the agent through the operation apparatus **41**. Here, the operation control unit **411** causes the agent to perform an operation of supporting the user through the operation apparatus **41**. Because the support that the operation control unit **411** causes the agent to perform is the same as that in the first embodiment, description thereof will be omitted.

(119) The position calculation unit **412** determines the position of the agent in the real space. Because an operation of the position calculation unit **412** is the same as that of the position calculation unit **312**, description thereof will be omitted.

(120) Next, an operation example of the agent system **100'** will be described.

(121) FIG. **13** is a sequence diagram illustrating an operation example of the agent system **100'**.

(122) Here, it is assumed that the speaker intends to start communication with the user.

(123) First, the spatial information management unit **413** outputs the speaker position information indicating the position of the speaker to the operation control unit **411** (S31).

(124) The operation control unit **411** calculates the angle p for rotating the agent based on the speaker position information from the spatial information management unit **413** (S32). When the angle p is calculated, the operation control unit **411** performs control for causing the operation apparatus **41** to rotate the agent by the angle p (S33).

(125) The operation apparatus **41** rotates the agent according to the control from the operation control unit **411** (S34).

(126) Here, the input apparatus **42** acquires the posture information of the user and the user position information (S35). When the input apparatus **42** acquires the posture information and the user position information, the input apparatus **42** inputs the acquired posture information and user position information to the information processing apparatus **3** (S36).

(127) When the input apparatus **42** inputs the posture information and the user position information, the spatial information management unit **413** outputs line-of-sight information indicating the vector indicating the line-of-sight of the user, the position of the user, and the like, object information indicating objects installed in the space, positions of the objects, and the like, the grid point information indicating the grid point in the relative coordinate system, and the like to the position calculation unit **412** (S37).

(128) The position calculation unit **412** determines the position of the agent based on the information from the spatial information management unit **413** (S38). When the position of the agent is determined, the position calculation unit **412** outputs the agent position information indicating the position of the agent to the operation control unit **411** (S39).

(129) The operation control unit **411** performs control for causing the operation apparatus **41** to move the agent to the position indicated by the agent position information from the position calculation unit **312** (S40).

(130) The operation apparatus **41** moves the agent according to the control from the operation control unit **411** (S41).

(131) The agent system **100'** may return to S31 or S35.

(132) In the agent system configured as described above, it is possible to place the agent at a position easily visible to the user while avoiding an area watched by the user, like the agent system of the first embodiment, in the real space.

(133) The present invention is not limited to the above embodiments, and can be variously modified at an implementation stage without departing from the gist thereof. For example, a type or configurations of the information processing apparatus, a type or configuration of display device, a procedure and content of information presentation position determination processing, a type or generation method of presented information, and the like can be modified and implemented in various ways without departing from the gist of the present invention.

(134) Further, the embodiments may be implemented through appropriate selection or a combination as much as possible. Further, the embodiments include inventions at various stages, and various inventions can be extracted by an appropriate combination in a plurality of disclosed constituent requirements.

REFERENCE SIGNS LIST

(135) **1** Headset **3** Information processing apparatus **11** Display apparatus **12** Input apparatus **31** Processor **32** Memory **33** Input and output interface **34** Communication unit **41** Operation apparatus **42** Input apparatus **100** Agent system **100'** Agent system **200** Virtual space **201** Avatar **202** Agent **203**

Speaker **204** Object **205** Object **206** Object **210** Agent placement area **311** Operation control unit **312**
Position calculation unit **313** Virtual space control unit **411** Operation control unit **412** Position
calculation unit **413** Spatial information management unit **414** Environmental information
acquisition unit

Claims

1. An information processing apparatus comprising: an interface configured to transmit or receive data to or from an input apparatus configured to input user position information indicating a position of a user; and a processor configured to; calculate an evaluation value for an evaluation target position based on the user position information, wherein the evaluation value is based on a function maximized at a value of a center between an angle of an end of a central visual field of the user and an angle of an end of a visual field of the user, determine a position in a virtual space or a real space where an agent supporting the user is placed, based on the evaluation value, and place the agent at the determined position in the virtual space or the real space.
 2. The information processing apparatus according to claim 1, wherein the evaluation value is based on a distance between the position of the user and the evaluation target position.
 3. The information processing apparatus according to claim 2, wherein the evaluation value is based on a function maximized at a center value between an upper limit and a lower limit of the distance.
 4. The information processing apparatus according to claim 1, wherein the input apparatus inputs posture information indicating a posture of the user, the processor calculates a vector indicating a line of sight of the user based on the posture information, and the evaluation value is based on an angle formed by a vector connecting the position of the user to the evaluation target position and the vector indicating the line of sight of the user.
 5. The information processing apparatus according to claim 1, wherein the evaluation value is based on the number of objects at a predetermined distance from the evaluation target position.
 6. An information processing method executed by a processor, the information processing method comprising: acquiring user position information indicating a position of a user; calculating an evaluation value for an evaluation target position based on the user position information, wherein the evaluation value is based on a function maximized at a value of a center between an angle of an end of a central visual field of the user and an angle of an end of a visual field of the user; determining a position in a virtual space or a real space where an agent supporting the user is placed based on the evaluation value; and placing the agent at the determined position in the virtual space or the real space.
 7. A non-transitory computer-readable medium storing program instructions that cause a computer to perform operations comprising: acquiring user position information indicating a position of a user; calculating an evaluation value for an evaluation target position based on the user position information, wherein the evaluation value is based on a function maximized at a value of a center between an angle of an end of a central visual field of the user and an angle of an end of a visual field of the user; determining a position in a virtual space or a real space where an agent supporting the user is placed based on the evaluation value; and placing the agent at the determined position in the virtual space or the real space.
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